



Business Research Methods

Topic:

Digital Loneliness

Submitted to:

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Contribution Table:

Fatima Farzand 2023-BBIT-10(A)	1- Figueiredo, E., Margaça, C., & Sánchez-García, J. C. (2025). Loneliness and Isolation in the Era of Telework: A Comprehensive Review of Challenges for Organizational Success. <i>Healthcare</i>, 13(16), 1943. https://doi.org/10.3390/healthcare13161943 2- Çolak, M. & Çetin, C. (2021). Loneliness and Cyberloafing in the Time of COVID-19: A Psychological Perspective. <i>International Journal of Contemporary Management</i>, 57(1), 2021. 15-27. https://doi.org/10.2478/ijcm-2021-0002
Ayesha Arshad 2023-BBIT-92(A)	1- Umdasch, M. C., & Berger, E. S. (2025). Understanding entrepreneurs' loneliness: An integrative review of loneliness in management research. <i>Journal of Small Business Management</i>, 1-40. 2- McCarthy, J. M., Erdogan, B., Bauer, T. N., Kudret, S., & Campion, E. (2025). All the lonely people: an integrated review and research agenda on work and loneliness. <i>Journal of Management</i>, 01492063241313320.

ABSTRACT

In the contemporary digital era, the widespread use of digital technologies and social media platforms has significantly transformed communication and social interaction. Despite increased online connectivity, many individuals experience digital loneliness, which refers to feelings of isolation and emotional disconnection even when actively engaged in digital environments. This quantitative study aims to examine the level of digital loneliness and to analyze its relationship with patterns of digital media usage, social interaction, and emotional well-being. Data were collected through a structured questionnaire designed to measure variables such as frequency of social media use, quality of online interactions, level of face-to-face communication, perceived loneliness, and the frequency with which individuals feel like quitting or withdrawing from social, academic, or digital engagement. Statistical techniques were applied to analyze the relationships among these variables. The findings indicate that excessive use of digital platforms is positively associated with higher levels of loneliness and emotional exhaustion. Individuals who frequently reported feelings of wanting to quit or withdraw also demonstrated elevated levels of digital loneliness, suggesting a strong connection between emotional disengagement and prolonged digital exposure. Furthermore, limited face-to-face interaction was found to intensify feelings of isolation. The study concludes that although digital technologies facilitate communication and accessibility, their excessive and unbalanced use may negatively affect emotional well-being and social connectedness. The report emphasizes the importance of balanced digital engagement and the promotion of meaningful offline interactions as effective strategies to reduce digital loneliness.

1-INTRODUCTION

In the digital era, professional work environments have undergone significant transformation due to increased reliance on digital technologies, remote work, and virtual communication tools. While these changes have improved efficiency and connectivity, they have also contributed to the growing issue of loneliness in professional life. Loneliness has emerged as a critical concern affecting both employees and entrepreneurs, influencing productivity, motivation, emotional well-being, and overall work performance. Rather than being limited to an individual emotional experience, loneliness is increasingly recognized as an organizational and structural phenomenon that shapes workplace relationships and leadership dynamics.

Previous research highlights that individuals spend a substantial portion of their daily lives at work, making professional environments a central context for social interaction (McCarthy et al., 2025). However, unmet social needs, reduced face-to-face interaction, and increased dependence on digital communication have intensified feelings of loneliness. The COVID-19 pandemic further accelerated this shift by normalizing remote work and digital collaboration, often at the cost of meaningful human connection (Ernst et al., 2022). Studies suggest that digital interactions may provide superficial connectivity without fulfilling deeper emotional and social needs (Wanberg et al., 2020).

Loneliness in professional contexts affects individuals emotionally, cognitively, and behaviorally, leading to frustration, anxiety, negative self-perceptions, social withdrawal, and reduced engagement (Heinrich & Gullone, 2006; Teneva & Lemay, 2020; Cacioppo et al., 2000). Entrepreneurs face additional challenges due to the independent nature of their work, lack of organizational support, and work–life imbalance, which further increase vulnerability to loneliness (Umdasch & Berger, 2025). Given these consequences, it is essential to systematically examine loneliness in work and entrepreneurial settings, particularly within increasingly digital environments.

1.1-RESEARCH GAPS

Despite growing academic interest in workplace and entrepreneurial loneliness, several important research gaps remain evident from the literature:

GAP 1:

Loneliness has been studied in workplaces and entrepreneurship, fewer studies explicitly examine loneliness within digitally mediated work environments.

GAP 2:

Most studies examine either employees or entrepreneurs separately, with limited comparative analysis between organizational and entrepreneurial contexts.

GAP 3:

Most studies examine either employees or entrepreneurs separately, with limited comparative analysis between organizational and entrepreneurial contexts.

1.2-PROBLEM STATEMENTS

PROBLEM STATEMENT 1:

The growing reliance on digital communication in workplaces has reduced direct social interactions, leading to digital loneliness that adversely affects employees' mental health, emotional well-being, and sense of organizational connectedness.

PROBLEM STATEMENT 2:

Digital loneliness can diminish employee engagement, motivation, and overall productivity, yet most organizations lack targeted strategies to address this challenge, highlighting the need to identify its causes and implement effective interventions.

1.3-RESEARCH OBJECTIVES

1- To examine the level of loneliness experienced in professional work and entrepreneurial settings within digital environments.

To analyze the relationship between digital work practices and feelings of loneliness.

1.4-RESEARCH QUESTIONS

RESEARCH QUESTION 1:

What is the relationship between digital work practices and the level of digital loneliness experienced by individuals in professional and entrepreneurial contexts?

RESEARCH QUESTION 2:

To what extent do feelings of withdrawal or the tendency to quit relate to emotional, cognitive, and behavioral effects of digital loneliness?

1.5-RESEARCH CONTRIBUTIONS

1.5.1- EMPIRICAL CONTRIBUTION:

This study provides quantitative evidence on digital loneliness among organizational employees by statistically analyzing the relationship between digital communication usage and feelings of loneliness.

1.5.2- ORGANIZATIONAL CONTRIBUTION:

The research extends organizational behavior literature by identifying how technology-based communication within organizations can influence employees' emotional connection and sense of belonging.

1.5.3- PRACTICAL CONTRIBUTION:

The findings help organizations design better digital communication strategies to reduce employee loneliness, improve engagement, and support employee well-being.

1.5.4- MEASUREMENT CONTRIBUTION:

By using a structured questionnaire, this study contributes by measuring digital loneliness in an organizational context, offering a reliable approach for future quantitative research.

2-LITERATURE REVIEW

2.1- Digital transformations in the workplace have reshaped how employees interact, communicate, and perform their duties (Çolak & Çetin, 2021). Employee psychological costs have increased along with additional flexibility opportunities brought about by the move to remote and hybrid work arrangements (Figueiredo et al., 2025). One of the arising psychosocial risks associated with this shift is **digital loneliness**, a kind of perceived social isolation that can occur even when workers are connected through digital platforms (Çolak & Çetin, 2021).

Understanding digital loneliness in organization is important because it hardly affects both individual well-being of employees and organizational outcomes such as productivity and knowledge sharing among employees (Figueiredo et al., 2025).

Loneliness in the digital workplace is **visualized** as a subjective feeling that results from a perceived mismatch between desired and actual social relationships at work (Çolak & Çetin, 2021). This subjective emotion can be triggered or intensified by digital modes of communication that lack the richness of in-person interaction (Figueiredo et al., 2025).

Researchers distinguish multiple dimensions of loneliness including **intimate**, **relational**, and collective forms all of which may be affected by digital work arrangements (Çolak & Çetin, 2021). The relational and collective aspects of workplace belonging may be weakened by the lack of spontaneous informal interaction and the low social presence in virtual channels (Figueiredo et al., 2025).

Many in-person contacts are replaced by digital technologies, but they frequently fall short of replicating the spontaneous conversations and emotional cues that maintain social ties (Figueiredo et al., 2025). When routine in-office interactions such as hallway conversations and coffee-break chats disappear, employees lose casual opportunities to connect, which increases feelings of detachment (Çolak & Çetin, 2021). The pandemic accelerated telework adoption and made remote collaboration normative, which in many cases increased perceptions of invisibility and reduced social support among employees (Figueiredo et al., 2025). Continuous online availability and the expectation of “always-on” can generate cognitive overload and social exhaustion, further spoiled interpersonal connection among employees (Figueiredo et al., 2025).

Employees who feel socially disconnected in digital work contexts may seek compensatory behaviors online, such as engaging in non-work internet activities during work hours (Çolak & Çetin, 2021).

Cyberloafing is positively associated with loneliness, indicating that online distraction may serve as a temporary coping mechanism for unfulfilled social needs (Çolak & Çetin, 2021). While cyberloafing may momentarily relieve feelings of isolation, it can also reduce task focus and productivity, creating a negative feedback loop for both workers and organizations (Çolak & Çetin, 2021). It is also noted that reduced idea exchange and lower participation in collaborative processes are additional behavioral outcomes linked to isolation in telework settings (Figueiredo

et al., 2025).

Contextual challenges arising from **COVID-19**, such as enforced remote work, the merging of home and work life, and increased technostress heightened feelings of loneliness and made the dynamics of cyberloafing more prominent across various industries. The pandemic also exacerbated role overload and conflicts between family and work for many remote employees, leading to greater emotional fatigue and a stronger inclination towards online, non-work-related activities as temporary distractions. As a result, cyberloafing during the pandemic often represented both coping mechanisms and structural pressures, rather than simply being a case of employee misconduct.

Loneliness has detrimental effects on one's health and long-term well-being. Prolonged loneliness increases the risk of depression, anxiety, and physical injury, which can result in **long-term productivity losses and turnover risks for businesses**.

Evidence even links loneliness to poorer sleep, cognitive deficits (attention and memory), and greater cardiovascular risk, underscoring the public-health dimension of workplace loneliness. Therefore, workplace loneliness should be considered both an individual mental-health issue and an organizational performance concern.

Persistent digital loneliness correspond with negative emotional states such as stress, anxiety, and **destroyed job satisfaction**, which in result increase the risk of burnout (Figueiredo et al., 2025). At the organizational level, loneliness erodes team cohesion and impairs knowledge sharing, thereby reducing innovation and the capacity for effective collective problem solving (Figueiredo et al., 2025).

It has highlighted how loneliness can lower engagement and motivation, which affects both individual performance and overall organizational productivity (Çolak & Çetin, 2021). The combined evidence suggests that digital loneliness should be treated as both a mental-health concern and a strategic business issue (Figueiredo et al., 2025).

Social Presence Theory helps explain why mediated communication often feels less satisfying than face-to-face contact, because digital channels convey fewer nonverbal and emotional cues (Figueiredo et al., 2025).

According to the **Conservation of Resources (COR) framework**, employees who are unable to refill emotional support in digital contexts experience stress and disengagement due to the

perceived loss of social resources (Çolak & Çetin, 2021). So, these theoretical lenses clarify why reductions in social richness and perceived resource loss make digital work environments especially vulnerable to loneliness (Çolak & Çetin, 2021).

According to the bibliometric results, there has been a significant increase in scholarly attention to loneliness connected to telework since 2020, which is indicative of a greater understanding of the psychological and organizational consequences of this phenomenon (Figueiredo et al., 2025). The review identifies loneliness as a mediating factor between mental health disorders and job performance, linking emotional disconnection to lower engagement and productivity (Figueiredo et al., 2025).

Similarly, the digitalization of workspaces has blurred the lines between personal and professional lives, increasing emotional strain and encouraging habits like **cyberloafing**, which is the use of the internet for non-work purposes during working hours (Figueiredo et al., 2025). Studies recommend organizational interventions aimed at restoring social connection, including structured virtual social activities and hybrid schedules that bring employees together periodically for in-person interaction (Figueiredo et al., 2025). Employees may be protected against loneliness by managerial strategies that prioritize psychologically safe communication, frequent check-ins, and emotional support (Çolak & Çetin, 2021). Practical strategies to lessen technostress and social isolation include training managers in compassionate leadership, digital wellness initiatives, and more defined responsiveness limits (Figueiredo et al., 2025). Implementing these strategies requires intentional design choices that prioritize social connectedness alongside digital efficiency (Çolak & Çetin, 2021).

It has been noted that much of the existing evidence derives from pandemic-era studies, creating a need for longitudinal research to examine the persistence and long-term effects of digital loneliness in post-pandemic hybrid work models (Figueiredo et al., 2025). There is also a shortage of cross-cultural and sectoral comparisons that would clarify how demographic, organizational, and cultural variables moderate loneliness outcomes (Çolak & Çetin, 2021). Finally, more experimental and intervention studies are needed to evaluate which organizational practices most effectively restore social presence and reduce the behavioral consequences associated with loneliness (Figueiredo et al., 2025)

Digital loneliness is a complex, multi-dimensional phenomenon that emerges where

technological connectivity replaces embodied social interaction (Çolak & Çetin, 2021).

Left unaddressed, it harms employee well-being and undermines organizational performance through lowered engagement, diminished knowledge sharing, and increased coping behaviors that distract from work (Figueiredo et al., 2025).

Mitigating digital loneliness requires combining technology design with human-centered management, including hybrid work models, manager training in empathetic leadership, and programs that rebuild informal social ties (Çolak & Çetin, 2021; Figueiredo et al., 2025).

2.2- Loneliness has become a major issue in professional life, which affects both employees and entrepreneurs. Its not like a problem at individual level but as a factor that shapes productivity, motivation, and well-being in organizations and business ventures. (McCarthy.,et al 2025) define loneliness as an important element in terms of work, where people spend there most of the time on daily basis. (Umdasch & Berger 2025) say loneliness among entrepreneurs has not been studied, even though it affects their health and work performance. They think loneliness isn't just an individual level problem but also an organizational level issue that impacts how people work and lead the team.

According to (McCarthy., et al. 2025),define loneliness as “a gap the social connections people have and the social connections they want.” (Peplau and Perlman 1982),said loneliness is when you want specific connections but the connections you have aren't the same. This difference makes you feel lonely at workplace . Similarly, (Baumeister and Leary, 1995) said that people need to feel connected for motivation and productivity. Without these connections, loneliness can arise. (Weiss ,1973) further defines that loneliness is very stressful feeling with no positive aspects.its a constant feeling of being alone.

(McCarthy et al., 2025) say that pandemic changed the things and loneliness increased at work and people feel more lonely. During, COVID-19 more people worked from home and more relied on digital tools for the communication purpose and face to face interactions has been reduced (Ernst et al., 2022). Studies shows that we aren't really connected online (Wanberg et al., 2020). Remote work can create more loneliness (Taser et al., 2022), and artificial intelligence can make us feel more isolated (Tang et al., 2023) .

Loneliness at work affects people in many ways: **emotionally, cognitively, and behaviorally**. Emotionally, it hurts people feeling and they are more frustrated and anxious and it also affects their whole day (Heinrich & Gullone, 2006). Cognitively, loneliness makes people more negative about others & themselves (Teneva & Lemay, 2020). Behaviorally, loneliness can make people stop participating in social activities (Cacioppo et al., 2000). McCarthy et al. (2025) This creates a cycle of feeling lonely and unhappy.

Two types of loneliness : **Transient loneliness**, temporary feelings that happen in certain situations, and **chronic loneliness**, permanent feelings of loneliness which persists over time (Cacioppo et al., 2006). Transient loneliness can motivate people to connect with others, like physical pain alerts of danger. However, chronic loneliness can make people stuck and with no hope (Maier & Seligman, 1976, 2016). Loneliness is different from: **social isolation** (Stephens, Shankar, Demakakos, & Wardle, 2013) not having social connections, **ostracism**, when people intentionally excluded or left out (Williams, 2007), and **depression** is a bigger mental health problem that can include feelings of loneliness (Erzen & Cikrikci, 2018).

The theoretical foundation for workplace loneliness can be seen from multiple perspectives. **Unmet social needs** can cause loneliness (Maslow, 1943), while **social exchange theories** say that poor interactions weaken social bonds (Lawler, 2001). Socio-psychological theories (Weiss, 1973) and **evolutionary approaches** (Cacioppo et al., 2015) helps people recognize and signals social disconnection.

(McCarthy et al. 2025) says workplace loneliness is complex and is influenced by **workplace structures, individual differences and group dynamics**. Organizations must address it with support, leadership and teamwork.

(Umdasch and Berger, 2025) studied loneliness in entrepreneurship — a negative emotional feeling due to their independent work nature (Ernst and Cacioppo, 1999) and (Peplau and Perlman, 1982). They face unique vulnerabilities because lack of structure, no coworkers, or no organizational support (Williamson et al., 2024).

However, recent studies (Cardon & Arwine, 2024; Zhu et al., 2023) say loneliness is a serious issue for entrepreneurs' mental health and business success. Loneliness can arise from limited

social support (Daniel et al., 2018). And work-life balance issues contribute to this loneliness(Mathias & Wang, 2023).

(Umdasch and Berger,2025) found two types of loneliness **state loneliness**: a temporary emotional experience and **trait loneliness**: a more stable condition. Entrepreneurs experience loneliness differently depending on their situation(Weiss et al., 1973). The authors says that entrepreneurs' loneliness develops over time. **Social isolation** : having few relationships (Holt-Lunstad et al., 2015), while **workplace isolation** defines lacking support from colleagues (Marshall et al., 2007). **Solitude**: can be positive when chosen for creativity or reflection (Long & Averill, 2003). But negative when involuntary or emotionally painful (Akrivou et al., 2011).

(Umdasch and Berger,2025) found that loneliness in entrepreneurs is complex, influenced by various factors, individual, social, and work factors. Personality, emotional intelligence, and work-life balance (Silman & Dogan, 2013). Social factors provides support and guidance (Gevaert et al., 2023), while excessive workload increase stress and obsessive passion can lead to burnout (Bayraktar & Jiménez, 2022). Future research use consistent definitions and valid measures, such as the UCLA Loneliness Scale (Russell et al., 1980) and the Loneliness at Work Scale (Wright et al., 2006), to improve comparability.

Both (McCarthy et al. 2025) and (Umdasch and Berger ,2025) agree that loneliness has multiple causes: Emotional, social, structural. Based on (Baumeister and Leary's,1995) belongingness theory, people need meaningful connections. In organizational settings, lack of inclusion or support, while in entrepreneurship, it results from independence and isolation. Both forms of loneliness share similar effects : emotional exhaustion, low motivation, and decreased creativity.

It also agrees that loneliness can be both temporary and chronic (Cacioppo et al., 2006). Support systems: such as coworkers, mentors, and family ,in coping mechanisms. (McCarthy et al. 2025) says team-building, as leadership empathy, and open communication as solutions, while (Umdasch and Berger,2025) says in peer networking, mentoring programs, and coworking spaces are for entrepreneurs. Employees have organizational support while Entrepreneurs must build their own networks (Galanti et al., 2021). Therefore, combating loneliness requires both organizational policies and personal agency for taking actions to build connections. Both can

benefit from emotional intelligence, self-awareness, and community engagement to reduce loneliness.

(McCarthy et al.,2025) suggests that loneliness as a serious well-being issue, like stress or burnout. Foster a culture of belonging through empathy, inclusion, and social connection which can help through emotional support. (Umdasch and Berger,2025) recommend that emotional support systems in entrepreneurship programs and incubators help founders to manage loneliness. Future studies should investigate that how technology, hybrid work(mixed work environments), and social media influence loneliness (Tang et al., 2023) and how tailor interventions to different contexts and needs.

In conclusion, both reviews shows that loneliness is a widespread challenge affecting people in offices ,start-ups and beyond in the digital era. the loss of human connection harms mental health and performance. Recognize the issue and its impact and learn ways to build meaningful connections and also create environments that foster belongingness. (Baumeister and Leary ,1995)said that, belongingness is a basic human need and modern work places must adapt to support it.

3-THEORATICAL FRAMEWORK

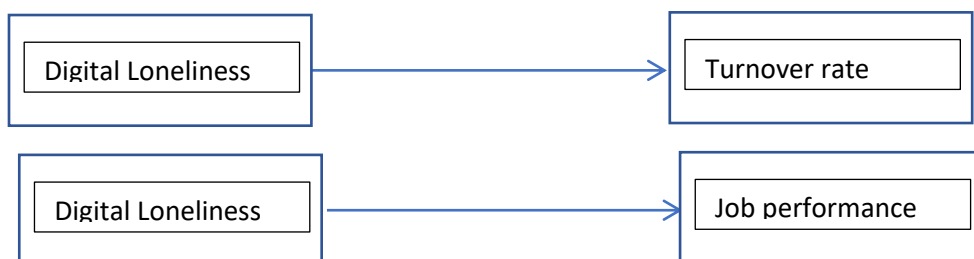
3.1-VARIABLES:

1-Independent Variable (IV):

Digital Loneliness is an independent variable in this study.

2-Dependent Variable(DV):

Employee Job Performance, turnover rate and burnout are dependent variables in this study.





3.2-HYPOTHESIS

- 1- Digital loneliness has a positive and significant effect on employee burnout.
- 2- Digital loneliness has a negative and significant effect on employee job satisfaction.
- 3- Digital loneliness has a negative and significant effect on employee job performance.

Hypothesis one or two are Directional Hypothesis as they show the direction of relationship(positive) among variables(digitalization and work flexibility,work efficiency).

Hypothesis number three is an Alternate Hypothesis as it shows there exist a relationship(significant influence) among variables(digitalization and employee well-being).

4-RESEARCH DESIGN

4.1-RESEARCH PHENOMENON:

The study explores a socio-psychological research phenomenon within organizational contexts. It is primarily psychological because it investigates employees' experiences of digital loneliness, emotional well-being, and mental health that emerge from extensive use of digital communication tools at work. Simultaneously, the phenomenon is social in nature because digital communication practices in organizations influence employees' social interactions, relationships, and sense of connectedness with colleagues.

4.2-TYPE OF RESEARCH :

i-On Application Basis:

The study is applied research because it intends to solve real world organizational problem. The research findings are meant to assist organizations , policy makers and managers in optimizing digital workplace services instead of developing theories only.

ii-On Objective Basis:

The research is explanatory because it seeks to explain the cause-and-effect relationship between digital communication practices (independent variable) and employees' digital loneliness, well-being, and productivity (dependent variables).

iii-On Time Dimension Basis:

The study adopts a cross-sectional research design, as data is collected at a single point in time from employees working in digitally intensive organizational environments. This design is suitable for understanding current perceptions and experiences of digital loneliness without tracking changes over time.

iv-On Inquiry Mode Basis:

The research follows a quantitative mode of inquiry since it relies on numerical data collected through structured questionnaires, primarily using Likert-scale items to measure levels of digital loneliness, emotional well-being, and work performance.

4.3-POPULATION:

The whole IT industry of Pakistan is the population in this research. The IT industry is highly digitalized that's why it is highly suitable for analyzing the impact of screen time, digital tools and hybrid work practices on employee mental health and work efficiency.

4.4-SAMPLE:

The IT industry across Lahore, Pakistan is selected as the sample for this study. Lahore is chosen because it is one of Pakistan's largest IT hubs, hosting numerous software houses, technology firms, and IT departments. This population is highly suitable for analyzing the impact of digital work practices, screen time, and digital communication tools on employees' experience of digital loneliness, mental health, and work efficiency.

4.5-SAMPLE SIZE:

Formula:

Where:

S = sample size

Number of items in questionnaire = 4

$n = 10 \times 4 = 40$

Therefore, the sample size is 40 respondents..

4.6-SAMPLING FRAME:

The sampling frame includes registered IT companies in Lahore, employees working in software houses or IT departments, and professional networks such as LinkedIn.

4.7-SAMPLING TECHNIQUE:

The study uses a non-probability purposive sampling technique. This technique is employed to select participants who directly interact with digitalized work environments, so they can meaningfully assess the impact of digitalization on digital loneliness, mental health, and work efficiency.

4.8-TYPES OF DATA :

The study uses primary data, as information is collected directly from IT employees regarding their current perceptions, experiences of digital loneliness, social interaction, and overall well-being in digitalized workplaces. This data is then used for statistical analysis, hypothesis testing, and answering research questions.

4.9-DATA COLLECTION METHOD:

Data is collected using a structured, close-ended research questionnaire. This method is selected because, in quantitative research, questionnaires allow standardized data collection and support statistical testing through SPSS.

4.10-RESEARCH RESPONDANTS:

The research respondents include software developers, IT engineers, IT project managers, and HR managers of IT firms. These employees actively interact with digital tools and platforms—for example, HR managers use digital systems for recruitment, payroll, and attendance—making them suitable for assessing the effects of digitalized work environments on digital loneliness, well-being, and work performance.

4.11-METHOD OF DATA ANALYSIS:

Data will be analyzed using Statistical Package for the Social Sciences (SPSS) through descriptive statistics and regression analysis. This software allows testing relationships between variables, measuring the strength of effects, and evaluating the impact of digital work practices on digital loneliness, mental health, and employee productivity.

5-ANALYSIS

5.1-DESCRIPTIVE FREQUENCY ANALYSIS:

Descriptive analysis is used to summarize the demographic attributes of respondents based on data collected through questionnaires and analyzes using SPSS. It is used to identify how many respondents responded to all of the questionnaire items and how many of the respondents missed it. The analysis helps in determining the variance among the level of agreement or disagreement among respondents with regard to each questionnaire item. Following is the descriptive analysis profile of respondents :

Statistics						
		Age	Gender	Working Experience	Role/Position	Knowledge transfer can be optimized through digitalization
N	Valid	52	52	52	52	52
	Missing	0	0	0	0	0

Statistics

		Communicati on processes can be optimized through digitalization	Documentatio n can be unified/simpli fied through digitalization	Digitalization allows work information materials to be designed in a new and clearer way	Employees' digital skills can be strengthened	Collaboration becomes more flexible through digitalization
N	Valid	52	52	52	52	52
	Missing	0	0	0	0	0

Statistics

		Employees support WHP project on digitalization	Analog processes become more efficient through digital support	Workplaces are made more user- friendly (e.g., more efficient equipment)	Managers recognize the potential of digitalization for health	Time management can be improved through digitalization
N	Valid	52	52	52	52	52
	Missing	0	0	0	0	0

Statistics

		Employees advocate digitalization	Creating awareness of the benefits/harm s of social media	Creating awareness for dealing with violence on the Internet	Work can be made more flexible (e.g., home office)	Employees work more independently through digitalization
N	Valid	52	52	52	52	52

Missing		0	0	0	0	0
Statistics						
		Digital communication simplifies collaboration (e.g., meetings via Skype)	Priority setting is facilitated by digitalization	Stress and psychological strain can be reduced through digitalization	Exercise/healthy nutrition/healthy lifestyle of employees can be promoted through digital offers	
N	Valid	52	52	52	52	
	Missing	0	0	0	0	

Table 5.1**5.2- REGRESSION ANALYSIS:**

Regression analysis is used to show the impact of independent variable (digitalization) on dependent variables (employee well-being, workplace efficiency, and flexibility). It helps in determining the strength and direction of relationships among variables. The purpose of using regression analysis is to analyze whether selected independent variables clearly indicates dependant variables.

Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	Digitilization _IV ^b	.	Enter

a. Dependent Variable: DV

b. All requested variables entered.

Table 5.2(a)

Model Summary^b

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.804 ^a	.647	.640	3.476

a. Predictors: (Constant), Digitilization_IV

b. Dependent Variable: DV

Table 5.2(b)

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1106.527	1	1106.527	91.601	.000 ^b
	Residual	603.992	50	12.080		
	Total	1710.519	51			

a. Dependent Variable: DV

b. Predictors: (Constant), Digitilization_IV

Table 5.2(c)

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	3.333	1.666		2.001	.051
	Digitilization_IV	1.538	.161	.804	9.571	.000

Coefficients^a

95.0% Confidence Interval for B

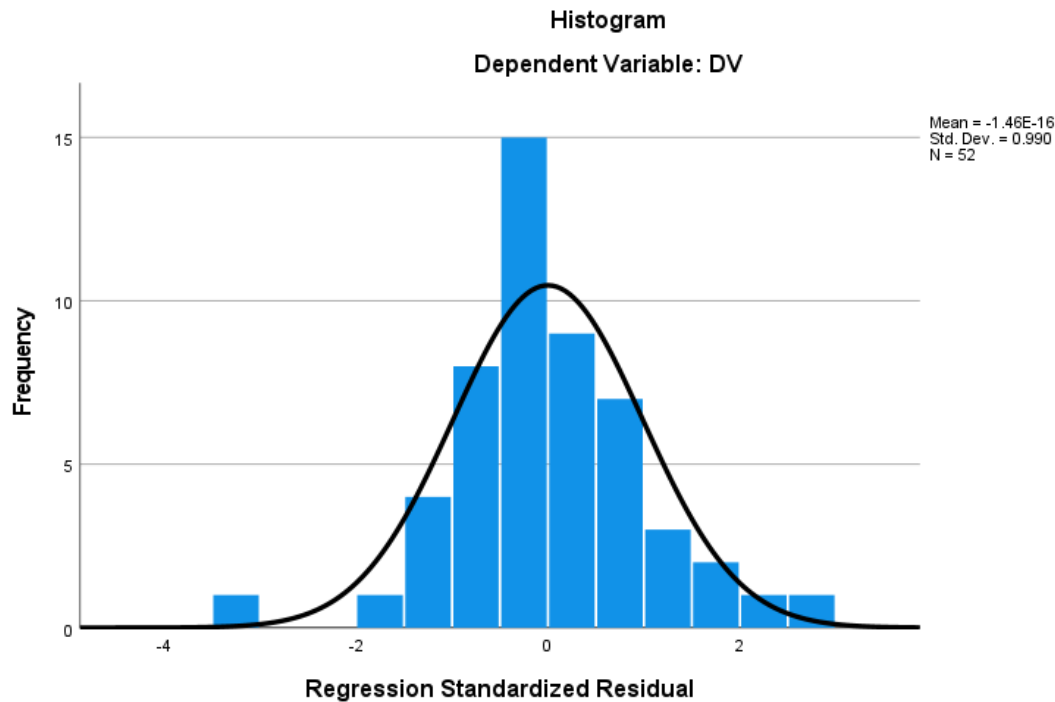
Model		Lower Bound	Upper Bound
1	(Constant)	-.013	6.679
	Digitilization_IV	1.215	1.861

a. Dependent Variable: DV

Table 5.2(d)Residuals Statistics^a

	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	14.10	34.10	18.60	4.658	52
Residual	-10.943	10.209	.000	3.441	52
Std. Predicted Value	-.965	3.328	.000	1.000	52
Std. Residual	-3.149	2.937	.000	.990	52

a. Dependent Variable: DV

Table 5.2(e)**Figure 5.1**

6-INTERPETATIONS

The Table 5.1 shows that all questionnaire items receives valid response without missing an item , the respondents participated actively in survey and answered all questionnaire items related to digitalization, employee well-being, workplace efficiency, and flexibility .The Table 5.2(a) represents that Digitalization-IV is entered as independent variables to investigate its impact on dependent variable. It indicates that no variable is removed from model as all variables met the specified criteria. The enter method in table shows that all independent variables are added into the regression model simultaneously.The Table 5.2b) shows the overall strength of regression model. The value of $R=0.804$ represents a strong positive correlation among digitalization and all the dependent variables. It indicates that change in digitalization strongly impacts employee

outcome. The Table 5.2(c) represents the the significance value =0.00 , which is less than the threshold value i.e (0.05). Since p value is less than 0.05 , it indicates that the alternate hypothesis and directional hypothesis are accepted i.e digitalization has significant impact on workplace efficiency, flexibility and employee well-being. The results suggests that the use of digital tools help employee manage and perform their task more efficiently and flexibly while maintaining good psychological well-being. It also supports existing literature that emphasized productivity and work-life balance at digital workplaces when digitalization is administered effectively.

7-CONCLUSION

This study concludes that digitalization has a significant impact on employee well-being, workplace efficiency, and flexibility. The regression results show a strong relationship between digitalization and the dependent variables, with $R = 0.804$ and a significance value of 0.000, which is below 0.05. The findings suggest that effective use of digital tools can improve employees' work efficiency and flexibility while supporting their psychological well-being. Therefore, organizations should adopt digitalization in a balanced and effective way to achieve better workplace outcomes.

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APPENDIX

Item	Mean	SD
1. I avoid taking decisions about integrity and security of my personal data online	2.38	1.04
2. When I am online, I often bookmark content (e.g., products I think of buying, content I would like to read) but never visit them later	2.26	1.22
3. When I am online, I avoid reading information that could cause unpleasantness	2.32	1.22
4. I avoid checking my e-mails	1.24	0.64
5. I just read news from online outlets that share my values	1.99	1.06
6. I unfollow and/or unfriend people on social media, with whom I disagree	2.05	1.18
7. I check my social media accounts as soon as I have a couple of minutes free time	2.93	1.41
8. I often get surprised when I realize how much time I have spent online	2.63	1.35
DHIS average	2.27	0.62

The range was 1–5 for all items. Cronbach's alpha=0.64

Mean Importance of the Opportunities of Digitalization in Companies

Which opportunities through digitalization have also arisen in your company, and how important or unimportant are these for your company?

	n	M	SD
Knowledge transfer can be optimized through digitalization	65	3.5	0.6
Communication processes can be optimized through digitalization	66	3.5	0.6
Documentation can be unified/simplified through digitalization	66	3.5	0.7
Digitalization allows work information materials to be designed in a new and clearer way	63	3.5	0.8
Employees' digital skills can be strengthened	64	3.4	0.7
Collaboration becomes more flexible through digitalization	64	3.3	0.8
Employees support WHP project on digitalization	59	3.2	0.7
Analog processes become more efficient through digital support	59	3.2	0.8
Workplaces are made more user friendly (e.g., more efficient equipment)	59	3.2	0.9
Managers recognize the potential of digitalization for health	57	3.2	0.9
Time management can be improved through digitalization	64	3.2	0.8
Employees advocate digitalization	60	3.2	0.6
Creating awareness of the benefits/harms of social media	61	3.1	0.9

DIGITALIZATION FOR HEALTH AND WELL-BEING AT WORK

13

Which opportunities through digitalization have also arisen in your company, and how important or unimportant are these for your company?

Creating awareness for dealing with violence on the Internet	60	3.0	1.0
Work can be made more flexible (e.g., home office)	57	3.0	1.2
Employees work more independently through digitalization	61	3.0	0.9
Digital communication simplifies collaboration (e.g., meetings via Skype possible)	61	2.9	1.1
Priority setting is facilitated by digitalization	61	2.9	1.0
Stress and psychological strain can be reduced through digitalization	61	2.7	1.0
Exercise/healthy nutrition/healthy lifestyle of employees can be promoted through digital offers	58	2.6	1.0

Note. 4=very important, 3=rather important, 2=rather unimportant, 1=very unimportant